

Natasha C. Greenstein

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EDUCATION

Universitat Pompeu Fabra
Master's in Brain and Cognition

Enrolled: Sep. 2025 — Exp: Jul. 2026

Princeton University
Bachelor's in Physics — Minor in Chinese Language and Culture
President Terrace F. Club (2024), President Princeton Science of Psychedelics Club, Taekwondo Team

Enrolled: Aug. 2021 — Graduated: May 2025
GPA: 3.3

RESEARCH EXPERIENCE

Quantum MemBrain Programme
Cofounder and Physicist

International
Jan. 2026 - present

- Integrating membrane dynamics as key player in multilayer resonance chain of biosignaling from quantum to planetary scales. Formalizing radical pair mechanism in mitochondrial electron transport chain and proton tunneling in ion channels as quantum origins.

Center for Brain and Cognition
Master's Researcher

Barcelona, Spain
Sep. 2025 - present

- Investigating whole brain fMRI activity as scale-invariant system whose multifractal behavior originates with the quantum mechanical wavefunctions of neurotransmitter bound receptors. Advised by Gustavo Deco.

Institute of Noetic Sciences
Physicist and Experimental Designer

Remote
Jul. 2025 - present

- Designing formalism and experiments linking changes in quantum information to changes in conscious states via agent interaction with quantum random number generators. Personally recruited by IONS President Thomas Brophy.

Princeton University
Bachelor's Researcher

Princeton, NJ
Sep. 2024 - Jul. 2025

- (I) Derived original Hamiltonian for radical pair mechanism in the context of bird migration - where biological effects are generalized as random Gaussian fields for general biological compatibility. Advised by William Bialek
- (II) Modeled human cortical activity with formalism of quantum mechanics: a Quantum-Like (QL) system. Verified compatibility of QL models with current neurophysiological models. Advised by Gregory Scholes

Max Planck Institute for Dynamics and Self-Organization
Neurophysics Intern

Göttingen, Germany
Jun. 2022 - Aug. 2022

- Modeled cerebrospinal (CSF) flow in mouse brain 3rd ventricle to compare CSF flow to phenotypical expression.

National Cancer Institute of NIH
Dosimetry Intern

Bethesda, MD
Jun. 2017 - May. 2020

- Numerically simulated radiation absorbed by medical phantoms. Analyzed spinal anatomy data from CT images. Constructed medical segmentations of CT scans for 2 projects. Performed Monte Carlo calculations. Wrote abstract.

PUBLICATIONS

Quantum-Like Dynamics in the Human Brain
Preprint on bioRxiv

Princeton, NJ and Barcelona, Spain
Jan. 2024 - present

A novel method for rapid estimation of active bone marrow dose for radiotherapy patients in epidemiological studies

American Association of Physicists in Medicine 2024

Bethesda, MD
Jul. 2019 - May 2020

Investigation of the influence of thyroid location on iodine-131 s values
Radiation Protection Dosimetry 2020

Bethesda, MD
Jun. 2017 - Aug. 2019

INVITED TALKS

From spin states to neural correlates: the quantum mechanical basis of avian navigation
Barcelona Computational, Cognitive and Systems Neuroscience (BARCCSYN)

Girona, Catalunya, Spain
Jan. 2026

FELLOWSHIPS

Science Outreach Fellow

Fellow

Princeton, NJ
Sep. 2023 – May 2024

- Developed and executed science outreach programs with the surrounding community to advance scientific education. Live physics demonstrations in schools. 1 of 20 students selected.

PROJECTS

Quantum Anthropology

Physicist

Remote
Jan. 2026 – present

- Formalizing, simulating, and investigating the role of a radical pair mechanism in human cryptochrome underlying visual aspects of psychedelic experience with Dr. Rebekah Senanayake.

Entangled Labs

Analyst

Remote
Dec. 2025 – present

- Analyzing output from Entangled Labs quantum random number generators, examining correlations with astronomical information, such as moon phase.

EMPLOYMENT AND VOLUNTEER WORK

Freelance Tutor

Physics and Mathematics

Remote
Sep. 2018 – present

Princeton Physics Tutor

From Classical to Quantum Mechanics (PHY207), Introduction to Quantum Mechancis (PHY208), Thermal Physics (PHY301)

Princeton, NJ
Sep. 2023 – May 2025

Taekwondo Instructor

Third degree black belt, founder of Open Door Taekwondo

Bethesda, MD and Princeton, NJ
Sep. 2018 – present

House of Milae

Waitress, social media manager

Bethesda, MD
Jun. 2021 – present

AWARDS AND SKILLS

- **Awards:** Senior Thesis Funding (2024), International Internship Program Award (2022), Youth Changemaker Award (2020), Bausch and Lomb Honourary Science Award (2019)
- **Relevant Coursework:** Biophysics (Grad), Quantum Field Theory (Grad), Quantum Information and Entanglement (Grad), Cosmology, Advanced Quantum Theory, Thermal Physics, Biophysics, Quantum Mechanics, Introduction to Quantum Computing, Fission and Fusion Energy, Classical to Quantum Mechanics, Integrated Science Curriculum
- **Languages:** Mandarin Chinese (fluent), Spanish (proficient)
- **Technical:** AI communication, MATLAB, python, Fourier analysis
- **Character traits:** Adaptable, creative, intellectually curious, experimental, investigative, disciplined, intentional, analytical, perceptive, imaginative